

2.4.7

EE25BTECH11017 - Mahesh Chollangi

Question Find the projection of the vector $\mathbf{a} = 2\hat{i} + 3\hat{j} + 2\hat{k}$ on the vector $\mathbf{b} = \hat{i} + 2\hat{j} + \hat{k}$.
Solution:

Let us find the solution theoretically first and then verify it computationally.
 Let the vectors be represented as $\mathbf{A}$, $\mathbf{B}$, and respectively. Given by

$$\mathbf{A} = \begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \quad (0.1)$$

Mathematically, the projection of the vector $\mathbf{A}$ on the vector $\mathbf{B}$ is given by some vector $\mathbf{D}$,

$$\mathbf{D} = k\mathbf{B}, \text{ such that } (\mathbf{A} - \mathbf{D})^T \mathbf{D} = 0 \quad (0.2)$$

yielding,

$$(\mathbf{A} - k\mathbf{B})^T \mathbf{B} = 0 \quad (0.3)$$

or,

$$k = \frac{(\mathbf{A})^T \mathbf{B}}{\|\mathbf{B}\|^2} \implies \mathbf{D} = \frac{(\mathbf{A})^T \mathbf{B}}{\|\mathbf{B}\|^2} \mathbf{B} \quad (0.4)$$

On substituting the values,

$$\mathbf{D} = \frac{\left(\begin{pmatrix} 2 \\ 3 \\ 2 \end{pmatrix} \right)^T \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}}{\left\| \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \right\|^2} \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \quad (0.5)$$

On calculation, this gives us,

$$\mathbf{D} = \begin{pmatrix} 5/3 \\ 10/3 \\ 5/3 \end{pmatrix} \quad (0.6)$$

We get the same result by plotting a graph for the following.

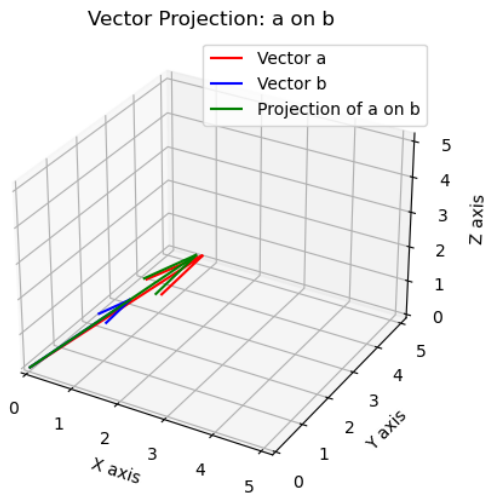


Fig. 0.1: 3D Plot